

MINISTRY OF HIGHER EDUCATION & SCIENTIFIC RESEARCH · REPUBLIC OF IRAQ

Reconstruction & Projects Department · دائرة الإعمار و المشاريع

Engineering Projects Management System

Technical Proposal for Department Management

Completing a system whose foundation is already built and verified.

All fourteen business rules are implemented and passing 128 automated tests against the department's own worked examples. One of twenty-four screens is complete. This proposal covers the remaining twenty-three, on the v1.1 design baseline.

Prepared for	Department management
Date	10 August 2026
Version	1.0 — for approval
Design baseline	v1.1 design system (<code>epm@design/system-revamp</code>)
Estimate	181–255 engineer-days · 6–8 months with two engineers
Decisions required	11 — eight business, three design

1 · Executive summary

EPM is the system of record for how the department contracts, builds and pays for construction and equipment projects across Iraq's public universities. It replaces spreadsheets and paper files for the department's highest-risk process: أمر الغيار — the change order that legally modifies a contract's value and duration.

14 / 14

Business rules built & tested

128

Automated tests passing

1 / 24

Screens complete

6–8

Months to completion

What we are asking for

Approval to complete the system on the delivery plan in §7, and decisions on the eleven items in §11. Several of those directly determine contract values and cannot be settled by engineering judgement.

Where the project stands

The foundation is complete and verified: database, API, shared component library, and — most importantly — **all fourteen business rules implemented as tested code**, with the department's own worked examples passing as automated tests.

What remains

Twenty-three screens across five phases. The critical path is **Contract → BOQ → Schedule → Change Orders**, which is sequential by nature: each layer computes from the one beneath it.

Estimate

181–255 engineer-days. With two engineers, approximately **six to eight months**. Adding a third engineer compresses this by roughly three weeks and no more — the schedule is governed by dependency, not headcount (§8). This is a bottom-up estimate from a known deliverable list and should be re-validated against measured velocity after Phase 2.

Principal recommendation — adopt the v1.1 design system now, before the remaining twenty-three screens are built.

Adopting today requires re-verifying **one** screen. After Phase 3 it is nine screens plus the workspace shell; after Phase 6, more than twenty-five. The cost of this decision rises every week it is deferred.

2 · What the system does

The department administers many projects; each project runs several contracts; each contract carries hundreds of bill-of-quantity lines and a Primavera schedule. Changes to those contracts are frequent and consequential.

The system exists to prevent five specific, recurring failures.

Failure today	What the system guarantees
Nobody can state what a contract is actually worth — original award, applied amendments and approved-but-unapplied orders get conflated	One defensible, derivable contract value. Approvals are never counted as commitments
Quantity changes get re-priced without authority	Up to 20% of the original quantity holds the original rate; only لجنة تثبيت الأسعار may price the excess

Approval chains are invisible — nobody knows who holds a file, or for how long	Every change order's position, owner and elapsed time, with automatic escalation
Physical progress is a number somebody typed	Progress computed from the schedule and rolled up by weight
History is lost when quantities are overwritten	Original, requested, approved and applied all persist and remain queryable

3 · Where the project stands today

Measured, not asserted.

Delivered	Evidence
Database, API and application skeleton	Runs; schema created on boot
All 14 business rules as tested code	128 automated tests passing , including the department's worked examples
Live rules reference	Every rule executes its worked example on request
Shared value lists — 20 enumerations, 117 entries	Verified: every stored code resolves to an Arabic and English label
Shared component library — 7 primitives	Built; 2 in production use
Projects list screen	Verified in browser, both languages, two breakpoints

Why finishing the rules first matters. The business rules are the part of the system where an error costs money. They are now fixed, tested and readable independently of any screen: a reviewer can open one file per rule, see the department's worked example, and see it execute. The remaining work is building screens on a settled foundation — a materially lower-risk activity than deciding rules and drawing screens at the same time.

4 · The design baseline — v1.1

This proposal takes the **v1.1 design system** as the visual and interaction baseline. That branch has been built, run and inspected; the screenshots that follow are of the working system.

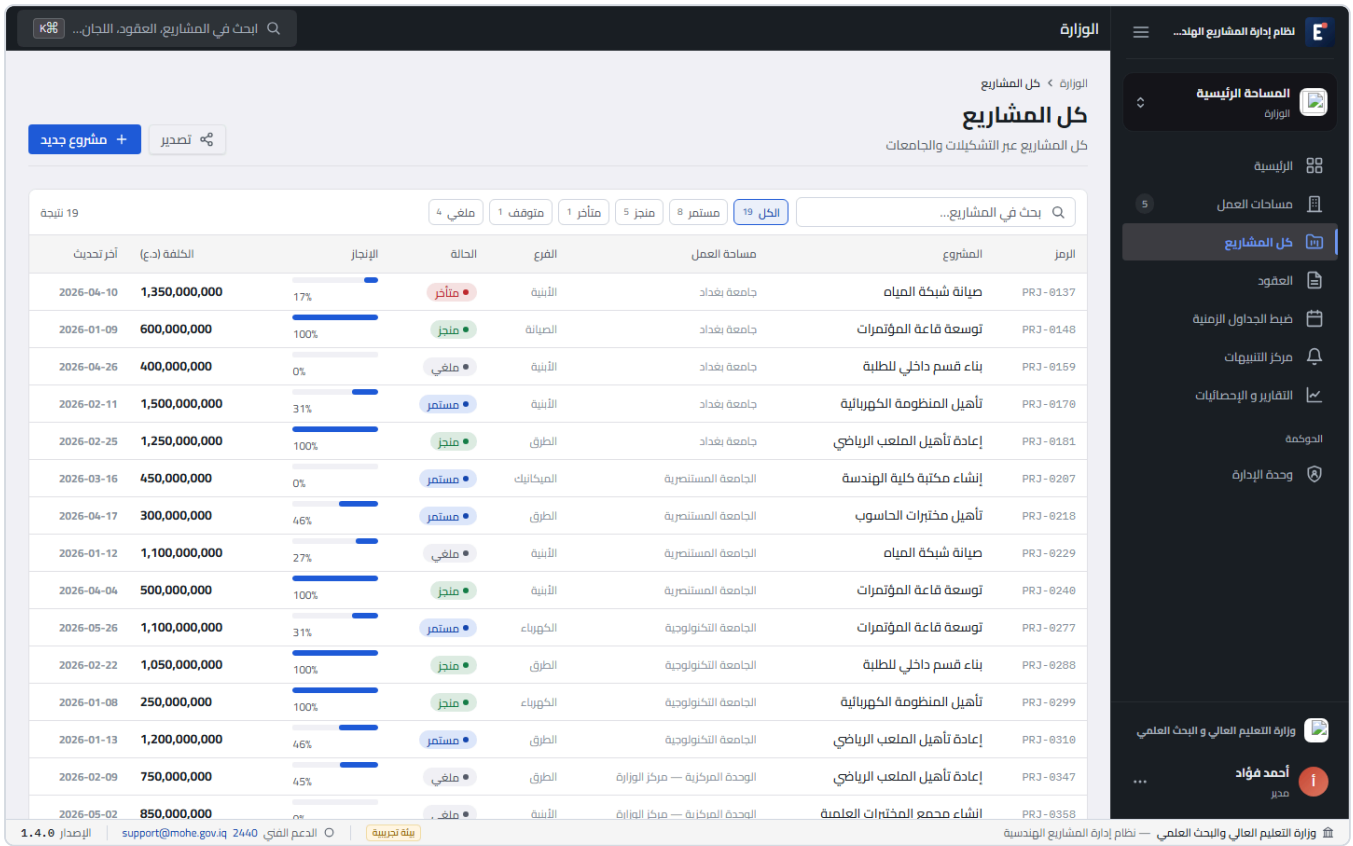
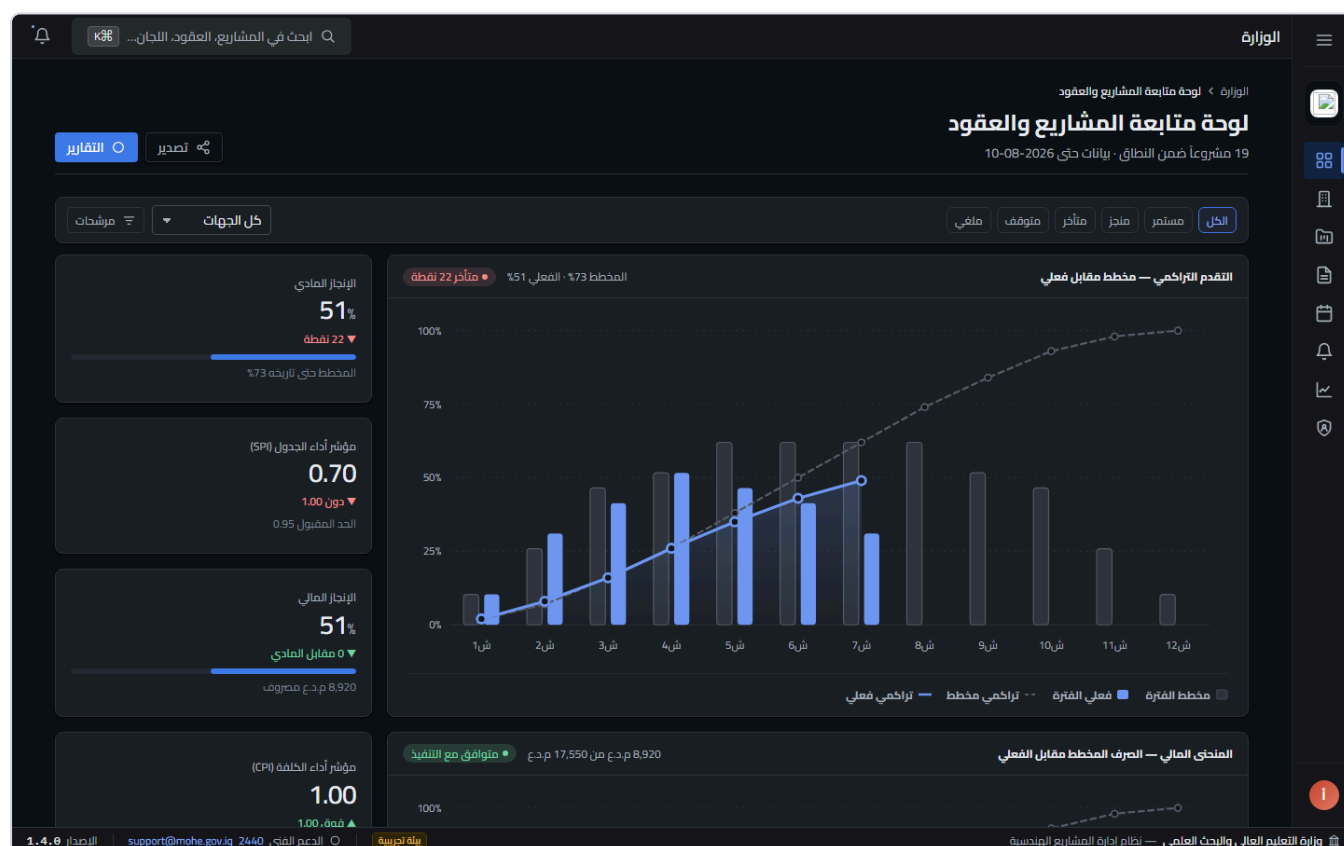
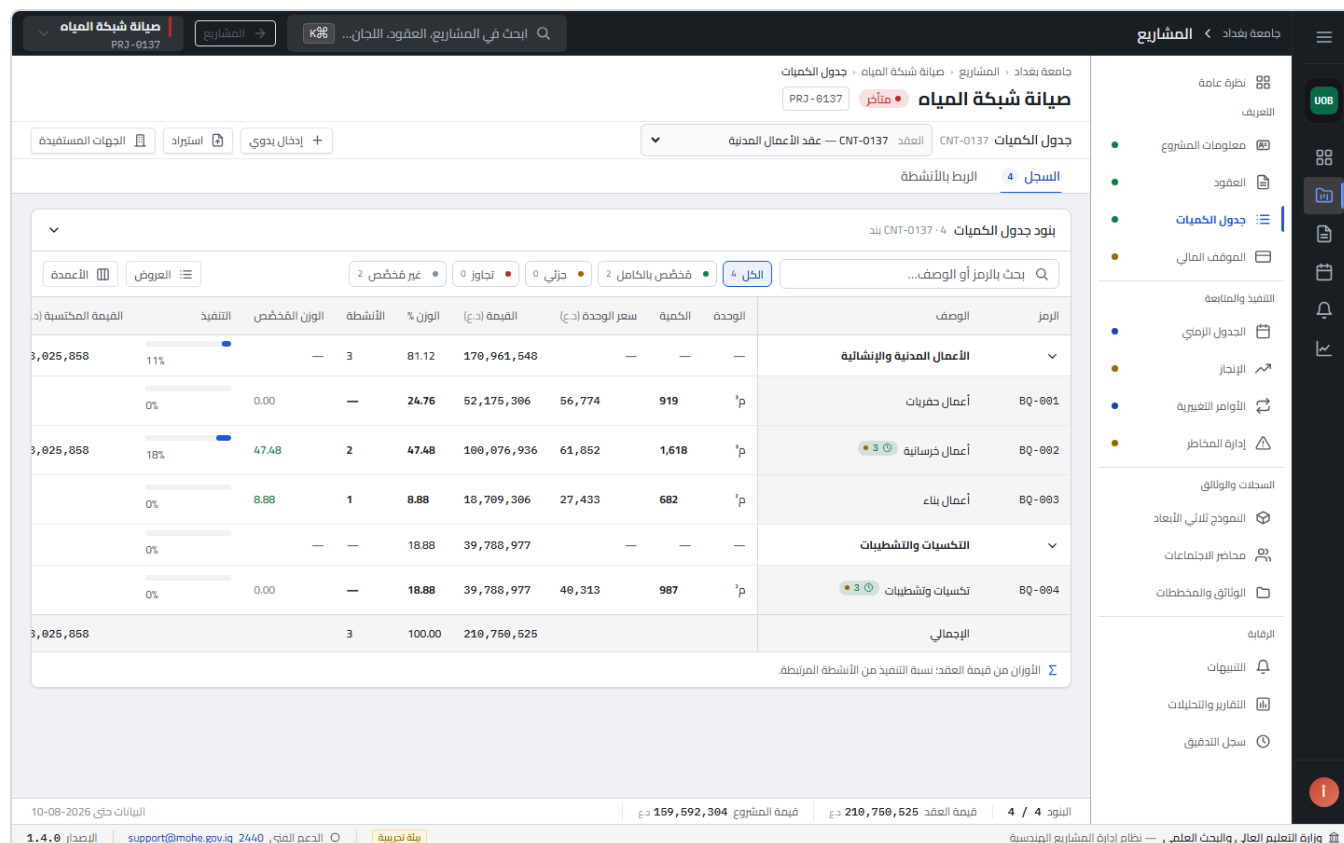


Figure 1 — The data-grid standard. Breadcrumb, page title with a single primary action, and a toolbar carrying search, filter chips and a live result count integrated into the grid card. Dense rows, status pills that always carry a label, right-aligned figures. Every register in the system follows this one pattern.



What v1.1 provides

Visual identity	Cool-grey canvas, EPM blue, Inter for Latin, Cairo retained for Arabic
Page architecture	A named zone contract every page follows: global bar → identity bar → toolbar → content → context panel
Data grid standard	Sticky header and identifier column, hierarchy with roll-ups, saved views, real pagination
Record standard	Header → tabs → grouped body → sticky footer, with a workflow rail and an audit tab
States	Skeleton, empty-with-action, error, permission-denied — specified, not improvised
Dark theme	Included
Rebuilt BOQ module	The department's densest screen, already designed and built

Why now — the cost of deferring

The migration was measured rather than estimated. v1.1 **re-skins the existing component contract rather than renaming it**: 50 of the 54 style classes the application uses survive unchanged, and every design token it references still exists.

Adopt at	Screens requiring re-verification
Now	1
After Phase 2	7
After Phase 3	9 + workspace shell
After Phase 6	25+

Adopting now also **de-risks the BOQ screen** — the hardest single item in the plan — because v1.1 has already designed and built it.

Two obligations that come with adoption.

- 1 • **The binding design specification becomes inaccurate.** It documents the previous palette and is the most-cited design document in the project. It must be re-baselined as part of adoption, not afterwards.
- 2 • **Two colour values fall below the accessibility floor.** The project's accessibility contract — the product of a formal audit — requires graphical elements to meet a 3:1 contrast ratio. Two v1.1 values measure **2.16:1**, where the previous values were chosen to sit at 3.03:1. They are used for chart baselines and the schedule's data-date line. The correction is small, but it must be a decision rather than a silent inheritance.

5 · Scope

In scope

Projects · contracts and amendments · bill of quantities · quantity distribution to beneficiaries · schedule and activities · progress · financials and payments · **change orders end to end** · risk · meetings · documents · alerts · reports · audit history · administration · bilingual Arabic/English throughout · role-based action gating.

Out of scope — agreed

Standalone claims management · quality (inspection, NCR, laboratory, material approval) · HSE · resource and ERP integration · GIS mapping · full mobile parity for newer modules · **real BIM/IFC rendering** (the tab is retained with a placeholder) · **real Primavera and Excel file parsers** (validation rules are built; file readers are not).

Authentication — deferred, and a hard blocker for production.

The system currently identifies users by a switcher, not a login. The permission *model* is real and enforced server-side; the *identity* is not. **No deployment carrying ministry data can proceed until real authentication is added.** It is excluded from the estimate below because its size depends on which identity provider the ministry mandates. This needs a decision now, and a separate estimate.

6 · Technical approach

Layer	Choice	Rationale
Web application	Angular 19	Arabic RTL and bilingual support are first-class
Server	.NET 9 minimal APIs	One file per feature; the queries behind a screen are readable in one place
Database	SQL Server	Ministry-standard; exact decimal arithmetic for money
Business rules	Isolated, framework-free layer	The specification as executable code

The one architectural rule

All business arithmetic lives in the rules layer. Screens and endpoints may filter, sort and display; they may not compute a weight, a price split, a penalty or an approval transition. This is enforced by review and is checkable automatically.

Money is never floating-point — the 20% price splits and percentage rounding must be exact to the dinar.

Traceability — the governance property

Every screen, endpoint, business rule and database table is indexed, and every endpoint shares a marker with the code that calls it. When a contract value is questioned, the path from the number on screen to the rule that computed it and the rows it read can be produced in minutes.

Documentation as code

Every business rule carries its specification text and the department's worked example. That example is simultaneously an automated test and is executed live on a rules-reference page. **A rule cannot be changed without its documentation visibly disagreeing with it.** This is working today.

7 · Delivery plan

Work proceeds one screen at a time, end to end — database, endpoint, screen, documentation and index entry — so nothing is built speculatively.

Phase	Contents	Depends on
0 · Foundation done	Platform, design system, shell, first screen	—
1 · Rules & primitives done	14 business rules + 128 tests, value lists, component library	0
1.5 · v1.1 adoption	Re-baseline the design spec, swap the system, add new page primitives, correct the two contrast values	1
2 · Enterprise screens	Contracts, Beneficiaries, Executive Portfolio, Alerts Centre, Schedule Control, Reports	1.5
3 · Workspace shell	Three-pane workspace, project detail, Overview, Project Information	1.5
4 · Business core	Contract & amendments → BOQ → Schedule → Progress & Financials	3
5 · Change orders	Register → record page → creation wizard → workflow & application	4
6 · Remaining tabs	Risk, meetings, documents, alerts, reports, audit, 3D placeholder	3
7 · Closeout	Rules documentation route, visual fidelity, responsive, accessibility and bilingual passes	5, 6

Contract before BOQ; BOQ before everything else. BOQ weights feed the allocation to schedule activities, which feeds progress, which feeds earned value and change-order impact. This ordering is a property of the business, not a preference.

8 · Effort, schedule, and the headcount ceiling

Estimate

Bottom-up, from the known deliverable list, sized by complexity tier. Each figure includes database, endpoint, screen, tests, documentation and browser verification in both languages, both themes and four screen widths.

Phase	Engineer-days
1.5 · v1.1 adoption	8 – 12
2 · Enterprise screens (6)	31 – 44
3 · Workspace shell (3)	18 – 26
4 · Business core (6)	46 – 65
5 · Change orders (4)	46 – 64
6 · Remaining tabs (7)	20 – 26
7 · Closeout	12 – 18

Total	181 – 255
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Excludes authentication (§5) and any scope arising from the eleven decisions in §11.

The headcount ceiling — the most important scheduling fact

Phases 3, 4 and 5 are **sequential**: each builds on the output of the one before it. Only Phases 2 and 6 parallelise usefully.

Team	Calendar estimate	Effect
1 engineer	9 – 12 months	Everything on one track
2 engineers	6 – 8 months	Second engineer absorbs Phases 2 and 6
3 engineers	5.5 – 7.5 months	Marginal — little for the third to do after Phase 2
4+ engineers	No further gain	Critical path is dependency-bound

Adding people will not compress this project materially beyond two engineers. Presenting a shorter date with a larger team would be misleading. What does compress the schedule is reducing scope, or resolving the §11 decisions early.

Confidence

- **Phases 1.5, 2, 6** — good confidence; similar to work already completed.
- **Phase 4 (BOQ, Schedule)** — moderate; the two densest screens. BOQ risk is reduced by v1.1 having built it.
- **Phase 5 (change orders)** — **widest uncertainty**. The most heavily specified area, and the one most exposed to the §11 decisions.

Recommend re-forecasting after Phase 2, when velocity on this codebase is measurable.

9 · Team

Role	Allocation	Responsibility
Full-stack engineer (lead)	Full-time	Critical path: workspace, BOQ, schedule, change orders
Full-stack engineer	Full-time	Enterprise screens, remaining tabs, closeout passes
Design / UX	Part-time	v1.1 conformance, screen review
Business analyst — department	Part-time, essential	Resolves \$11; validates against real contracts
QA	Part-time, from Phase 4	Acceptance, bilingual and accessibility passes

The business analyst is not optional. Eight of the eleven open items are questions only the department can answer, and each one blocks or reworks part of the change-order module.

10 · Quality and acceptance

Level	Coverage
Business rules	Every worked example from the specification, automated. Already passing: 128 tests
Integration	Applying a change order updates contract value, dates, quantities, weights and penalty
Permissions	Every role against every change-order state sees the correct available actions
End-to-end	Create → submit → return → resubmit → approve → apply → close, in Arabic
Accessibility	Automated audit on every screen against the binding contract
Bilingual	Every screen in both languages; no untranslated text, no misrendered figures

Definition of done for a screen: its table, endpoint, screen, documentation and index entry all exist and agree; all four data states handled; verified in the browser at 1440 and 1024, in both languages.

11 · Decisions required from management

Each of these will harden into contract values if left unanswered.

Business — the department must decide

#	Question	Current assumption	Consequence if wrong
1	When a second change order adds quantity beyond 20%, is the threshold measured against the original or the current quantity?	Original	Changes the price of every re-priced quantity after the first amendment
2	Is the delay penalty 0.1% per day, capped at 10% ?	Yes	Directly changes penalty amounts — must be checked against the contract template
3	What is the real time limit per approval stage?	5 days uniformly	Drives overdue flags and automatic escalation
4	May the committee rapporteur record a cancellation for any external body, or only two of them?	Only two	Determines who can terminate a change order
5	When a quantity decrease forces a distribution revision, who revises it ?	Undecided	Blocks the order from being applied
6	Which execution-stage list is correct? Two incompatible lists exist in the source material	The 12 construction stages	Wrong stage labels on every project
7	Two change-order states are missing from the data dictionary but required by the process	Added, with our Arabic wording	Without them, “approved ≠ applied” cannot be represented
8	Confirm the Arabic wording for the two states above	Ours	Terminology appears throughout the interface

Design and platform

#	Question	Recommendation
9	Adopt the v1.1 design system now?	Yes — now. Cost grows with every screen built
10	Correct the two colour values below the accessibility floor?	Yes. Small change; the contract is binding
11	Is the dark theme in scope?	Decide explicitly — it arrives free but doubles visual verification per screen

And one gating item. Authentication (\$5) is excluded from this estimate. The ministry must nominate the identity provider before production planning can complete.

12 · Risks

Risk	Likelihood	Impact	Mitigation
\$11 business decisions arrive late	Medium	High	Resolve items 1–5 before Phase 4; they are already documented with options

Change-order module underestimated	Medium	High	Re-forecast after Phase 2; sequence register → record → wizard → workflow so value lands incrementally
Design system adopted late or partially	High if deferred	Medium, rising	Adopt at Phase 1.5 as proposed
Accessibility contract quietly breached	Medium	Medium	Correct the two values at adoption; automated audit in the closeout pass
Primavera / Excel import expected in Phase 1	Medium	Medium	Explicitly out of scope; validation rules built, parsers not — confirm with stakeholders
Authentication treated as small	Medium	High	Decide the provider now; size separately
Real ministry data reveals rule gaps	Medium	Medium	Business analyst validates against real contracts from Phase 4

13 • Assumptions

1. The binding specification remains authoritative except where v1.1 supersedes the design document, which will be re-baselined.
2. The v1.1 branch is intended to become the reference of record. **It is currently unmerged and 27 commits ahead** — this should be confirmed before Phase 1.5 starts.
3. Demonstration figures are illustrative, not ministry data. Real data is required from Phase 4 for meaningful validation.
4. The department provides a business analyst for \$11 and for review at each phase exit.
5. Effort excludes authentication, migration from existing spreadsheets, production infrastructure, training and support.
6. Estimates are bottom-up from a known deliverable list, to be validated against measured velocity after Phase 2.

14 • Recommendation

1. **Approve the delivery plan** in §7 on a two-engineer team, targeting six to eight months for the scope in §5.
2. **Approve v1.1 adoption at Phase 1.5**, before the remaining twenty-three screens are built. This is the lowest-cost moment available and the cost rises every week.
3. **Correct the two accessibility values** as part of that adoption.
4. **Assign a departmental business analyst now** and resolve business decisions 1–5 before Phase 4 begins.
5. **Decide the authentication provider** and commission a separate estimate.
6. **Re-forecast after Phase 2**, when velocity on this codebase is measurable.

Supporting documents	
docs/SRS.md	Full requirements, business context, glossary
docs/PROPOSAL-design-system-v1.1.md	Technical detail of the design migration
ROADMAP.md	Phase-by-phase build checklist
DECISIONS.md	Every decision taken, with its reasoning
TRACE.md	Screen → endpoint → rule → table index
docs/spec/01 – 07	The binding specification